

# Enhancing Mu-based BCI Performance with Rhythmic Electrical Stimulation at Alpha Frequency

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**Abstract**—The accuracy of brain-computer interfaces (BCIs) is important for effective communication and control. The mu-based BCI is one of the widely used systems, of which the related methods to improve users' accuracy is still poorly studied. Here, we examined the way to enhance the mu-based BCI performance by rhythmic electrical stimulation on the ulnar nerve at the contralateral wrist at the alpha frequency (10 Hz) during the left- and right-hand motor imagery. Time-frequency analysis, spectral analysis, and discriminant analysis were performed on the electroencephalograph (EEG) data before and after the intervention of electrical stimulation in 9 healthy subjects. We found that the ERD/S on the somatosensory and motor cortex during left- or right-hand imagination was more obvious at the mu rhythm after intervention. Furthermore, average classification accuracy between left- and right-hand imagery significantly increased from 78.43% to 88.17% after intervention, suggesting that the electrical stimulation at alpha frequency effectively regulates the brain's mu rhythm and enhances the discriminability of the left-hand and right-hand imagination tasks. These results provide evidence that the electrical stimulation at the alpha frequency is an effective way to improve the mu-based BCI performance.

## I. INTRODUCTION

The brain-computer interface (BCI) provides a direct way for human to interact with the outside world through the power of thought [1]. The performance of a BCI system is important for the users to effectively communicate with the outside world. In these years, the research of BCI has made great progress. However, the BCI system is still not stable, which makes the popularity of this technology in daily application difficult. It is reported that currently 15-25% of users cannot control the BCI system through EEG signals [2,3].

In order to improve the usability of BCI system, many suggested to improve the accuracy to increase the user's ability to control the BCI. For example, it has been reported that more easily classified brain signals can be generated by improving the original paradigm [4], or by combining different paradigms [5] to enhance the classification accuracy. In addition, EEG signals that are easier to classify can be obtained using different stimulus schemes [6,7].

The motor-imagery based BCI is one of the most used BCI system. As we known, motor imagination is the mental simulation of kinesthetic movement, which can regulate mu rhythm of sensory motor cortex without physical movement of the body [8,9]. Therefore, the brain activity induced by left-hand and right-hand motor imagery is considered as a useful

feature of BCI [10]. In addition, due to the contralateral positioning of oscillatory activities, the activities caused by left-hand and right-hand motor imagery are relatively easy to distinguish [11-13].

In recent years, in order to enhance the discriminability of cortical activation in motor imagination process, many intervention methods have been used. Most of intervention methods were based on sensory, e.g., visual [14-16]. However, some participants did not produce more useful features in the sensorimotor cortex in the post-intervention than the features extracted in the pre-intervention [14,17,18]. The most typical reason for this phenomenon is that the subject is more inclined to visual imagination, which is a brain activation mode that is completely different from the actual motor imagination [19]. Therefore, in order to avoid the wrong motion imaging mode caused by sensory intervention as much as possible, this paper hopes to intervene through nerve intervention means to improve the classification accuracy of BCI.

For the nerve intervention, it has been reported that the repetitive transcranial magnetic stimulation (rTMS) at a specific alpha frequency used to enhance task-related alpha desynchronization [20]. Furthermore, the 10 Hz TMS intervention had been used to increase the accuracy of motor-imagery based BCI from 63.5% to 74.3% in motor-imagery tasks [6]. These suggest that external stimuli with a rhythm in the cerebral cortex can induce an increase in cerebral cortical rhythmic activity, resulting in an EEG signal that is easier to capture. In addition, we also found that the electrical stimulation on the ulnar nerve at a specific alpha frequency can enhance task-related alpha desynchronization [21]. Hence, we hypothesized that the use of 10 Hz electrical stimulation to intervene the ulnar nerve in the wrist area can enhance the task-related alpha desynchronization on the sensory motor cortex during the hand motor imagery, to improve the performance of mu-based BCI system.

## II. METHODS

### A. Subjects

Nine student subjects, including six men and three women, aged between 21 and 24 (average age  $22.2 \pm 0.89$ ), were recruited for the study. All nine participants in the study were right-handed and had good physical and mental health. No dyskinesia or mental illness were reported. All participants were naive to motor-imagery-based BCI that had not previously participated in any BCI experiments. The study was performed in accordance with the Declaration of Helsinki.

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## B. Experimental tasks and procedure

The experiment was carried out in a quiet and comfortable environment with natural ventilation and suitable temperature. Generally, it took 1.5 to 2 hours from wearing a conductive cap to the completion of all experimental trials. Participants were informed to sit in the comfortable and stable chair with their hands resting naturally on the armrests. There was a monitor 1.2 meters in front of the seat, which was used to guide the subjects to do motor imagery tasks. A conductive cap was wearing to record the EEG data.

First, before intervention (pre-intervention), an EEG data set was recorded during the left-hand and right-hand motor imagery tasks for the baseline. Then an intervention was conducted by rhythmic electrical stimulation on the ulnar nerve at the contralateral wrist at the alpha frequency (10 Hz) during the left- and right-hand motor imagery. At the end (post-intervention), an EEG data set was recorded again with the same paradigm as the pre-intervention for comparison.

### 1) Experimental design without intervention

The experimental design flow for a single trial is shown in Fig. 1:

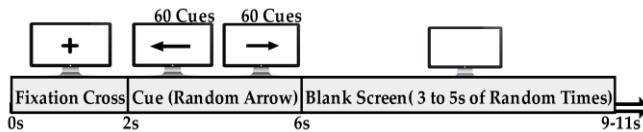


Fig.1. Paradigm for EEG data acquisition in a trial during motor imagery without electrical stimulation intervention.

Step 1, for the first 2 seconds, a fixed cross appeared in the center of the screen. At this stage, the subjects were in the preparation state without any motor imagination.

Step 2, From 2nd second, the cross at the center of the screen was changed to a black left or right arrow. When the left arrow appeared on the screen, the subject was asked to imagine the movement of the left hand up. While the right arrow appeared, the subject needed to imagine the movement of the right hand up. The subjects' imaginary motor tasks continued until the 6th second (the arrow disappeared from the screen). Facial activities such as blinking or limb were not strictly allowed throughout the mission.

Step 3, At the end of a single imagination task, the screen showed a random blank screen of 3-5s. This stage provided enough time for the subjects to complete blinking, swallowing saliva and other behaviors, so that the subjects can have interval rest to avoid fatigue in the task state.

The experiment of motor imagery included 120 trials, in which the left and right hands imagined 60 times each, and the order of the arrows were random.

### 2) Experimental design with intervention

The time course of a single trial during the experiment of intervention is the same as in Fig. 1. The difference was that during the motor imagery tasks, an electrical stimulation with 10 Hz was given to the ulnar nerve of the contralateral hand. For example, when the subjects imagined the task of raising their right hand/left hand, the electrical stimulation was given to the left-hand/right-hand ulnar nerve.

The left-hand and right-hand imagination task was performed separately. Each contained 30 single experimental trials. Participants took a break of 10 minutes between these two motor imagery tasks under intervention.

## C. EEG recording and preprocessing

EEG data were acquired from 64 scalp sites (extended 10-20 system) using an electrode cap (quick cap64). Reference electrode was bilateral mastoid. Through Neuroscan Synamps2 amplifier signal was amplified. The sampling rate of the EEG signal was 1 kHz. The band-pass filtering range was 0.5-30 Hz, and the electrode impedance was less than 5 K $\Omega$ . In the preprocessing stage of the data, this study used ICA to remove eye artifacts. EEG data from the 2nd to the 6th second of motor imagination tasks were extracted from all the trials. There were 120 episodes of EEG data to be extracted in pre-intervention, and in post-intervention separately.

## D. Signal Processing

### 1) Time-Frequency Analysis

In the process of motor imagination, motor sensory areas of the cerebral cortex were activated to varying degrees [11]. Therefore, EEG data from C3, C4 channels and their adjacent channels (C1, C2, C5, C6, CP1, CP2, CP3, CP4, CP5, CP6, FC3, FC4, FC5, and FC6) were selected to extract the power spectrum of mu-rhythm frequency band (8-13 Hz) at both pre-intervention and post-intervention of each subject in the 2nd to the 6th seconds. The power of 60 trials in the left-hand motion imagination and 60 trials in the right-hand motor imagination pre- and post- intervention was averaged. During the Time-Frequency Analysis, the resting state data before the task (1.5 to 2 seconds) was extracted as the baseline correction.

### 2) Spectral Analysis

For the motor imagery tasks before and after intervention, the data with a bandwidth of 5-30 Hz was extracted for spectrum analysis. The power average of all trials in the left-hand motor imagery or in the right-hand motor imagery were calculated separately.

### 3) Discriminant Analysis

This paper calculated the classification accuracy for pre-intervention and post-intervention of each subject separately. Data processing steps are shown below:

Step1, feature extraction. C3, C4 channels and their adjacent channels were selected to calculate the power of right and left imagination trials at mu-rhythm frequency band (8-13 Hz).

Step2, feature/channel selection. Due to the great individual difference in the activity of brain regions after intervention, we conducted significance analysis on these 16 features/channels of each subject to select the feature/channel with the most intervening. The EEG features/channels were selected with  $p < 0.0001$  of the T-test between pre-intervention and post-intervention for each subject to construct the characteristic matrix.

Step3, the calculation of classification accuracy. There were two categories: one was the left-hand motor imagery, and the other was the right-hand motor imagery. The classifier used the algorithm of support vector machine (SVM) and the final classification accuracy was obtained through 10-fold cross validation.

### III. RESULTS

The activation of the brain is usually contralateral during hand motor imagery. In order to present the results, C3 and C4 channels were selected as the main presentation.

#### A. Time-Frequency Analysis and Spectrum Analysis

Fig. 2 shows the energy changes of the C3 channel when a subject performs the right-hand motor imaging task. As mentioned in paper [22], the left-right hand motor imaging was obvious for the activation of the sensory motor areas, and it exhibits a contralateral activation mode. This paper also shows the same phenomenon which can be seen from the Fig. 2.

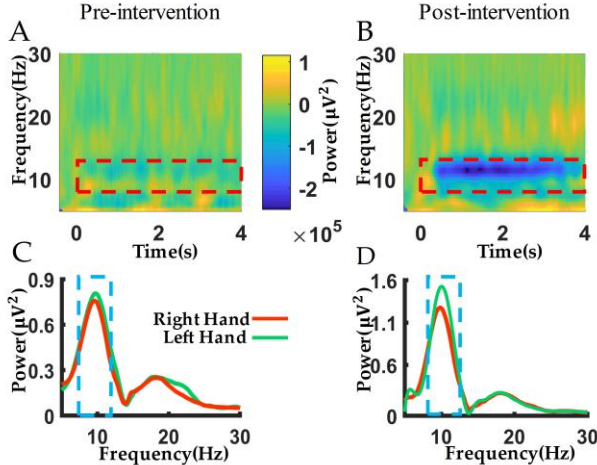


Fig. 2. The time-frequency analysis and spectrum analysis results in C3 channel. (A) The result of time-frequency analysis in pre-intervention. (B) The result of time-frequency analysis in post-intervention. (C) The result of spectrum analysis in pre-intervention. (D) The result of spectrum analysis in post-intervention. The power of (A) and (B) is calculated in millisecond. The power of (C) and (D) is calculated in second.

As seen in Fig. 2A and 2B, the power of the low frequency band (mu rhythm, 8-13Hz) was significantly reduced. The contralateral brain area (C3) showed obvious inhibition in post-intervention. However, no significant change was observed in the high frequency band (Beta band) before and after the intervention.

In the spectrum analysis of the C3 channel, the results were shown in Fig. 2C and 2D. We found that the power of the left-hand motor imagery is greater than the power of the right-hand motor imagery at around 10 Hz. This is because of the contralateral inhibitory effect in the mu rhythm during motor imagery tasks. Furthermore, we found that in the mu rhythm (8-13 Hz), the power value in post-intervention is significantly larger, and more importantly, the power difference between the left-hand and the right-hand motor imagery becomes larger. This suggests a greater degree of left-hand and right-hand motor imagery tasks separation, and suggests that left-hand and right-hand motor imagery tasks are easier to distinguish.

Fig. 3 shows the results of time-frequency analysis and spectrum analysis in C4 channel during the left-hand motor imagery. Like the results in channel C3 during right-hand motor imagery (Fig. 2), there was also significant contralateral inhibition and significant separation between left-hand and right-hand motor imagery tasks in the mu frequency band after intervention.

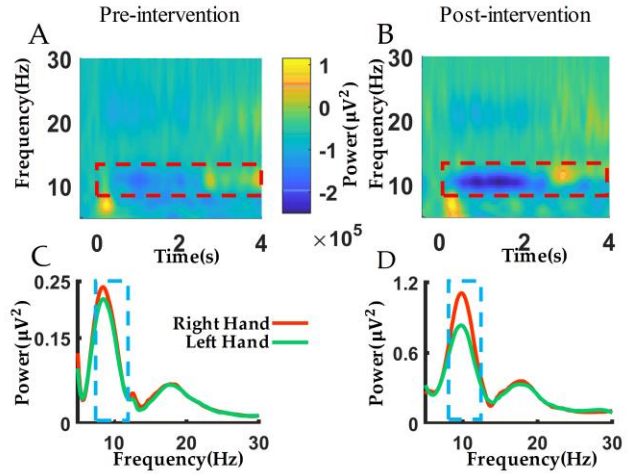


Fig. 3. The time-frequency analysis and spectrum analysis results in C4 channel. (A) The result of time-frequency analysis in pre-intervention. (B) The result of time-frequency analysis in post-intervention. (C) The result of spectrum analysis in pre-intervention. (D) The result of spectrum analysis in post-intervention. The power of (A) and (B) is calculated in millisecond. The power of (C) and (D) is calculated in second.

#### B. The results of average power

Fig. 4A is a summation average of several channels in the right brain region of 9 people when they imagined the left hand, while Fig. 4B is the summation average of several channels in the left-brain region of 9 people when they imagine the right-hand. There was a significant increase (T-test,  $P < 0.05$ ) after intervention in the average power during both left-hand motor imagery (Fig. 4A) and right-hand motor imagery (Fig. 4B). This indicates that the corresponding sensory motor cortex was significantly activated when performing left-hand or right-hand motor imagery task in the post-intervention.

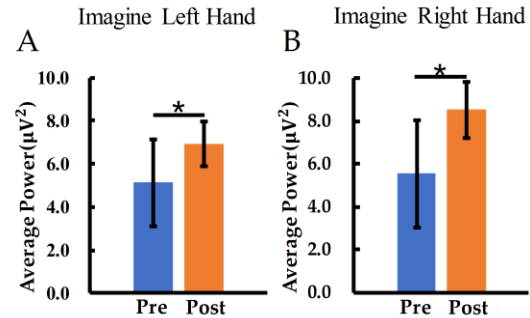


Fig. 4. Average power of 9 subjects. (A) Average power results in the right sensory motor areas when imagining the left hand in the pre- and post-intervention. (B) Average power results in the right sensory motor areas when imagining the right hand in the pre- and post-intervention. Pre is Pre-intervention, Post is Post-intervention.

#### C. Classification accuracy results

In Table I, we showed the classification accuracy results (%) of motor imagery tasks in the pre-intervention and post-intervention. There was almost no change in the results of pre-intervention and post-intervention in subject 4. However, the accuracy of the other subjects in post-intervention was much higher than that in pre-intervention. The average accuracy of all subjects was  $88.17 \pm 8.91\%$  in the post-intervention, while there was  $78.43 \pm 7.60\%$  in the pre-intervention. In order to further affirm this result, we performed T-test for the accuracy

of pre-intervention and post-intervention, and we found there is a significant difference ( $P = 0.002$ ). It is important to note that this reflects that the intervention with electrical stimulation (10 Hz) was effective for the discrimination between the left- and right-hand motor imagery tasks.

TABLE I ACCURACY RESULTS IN PRE-INTERVENTION AND POST-INTERVENTION.

| Subjects | Pre(%)     | Post(%)    |
|----------|------------|------------|
| Subject1 | 70.83      | 87.50      |
| Subject2 | 81.67      | 95.83      |
| Subject3 | 87.50      | 95.17      |
| Subject4 | 76.67      | 75.00      |
| Subject5 | 91.67      | 97.50      |
| Subject6 | 79.17      | 88.00      |
| Subject7 | 71.67      | 92.50      |
| Subject8 | 69.17      | 76.17      |
| Subject9 | 77.50      | 85.83      |
| Mean±SD  | 78.43±7.60 | 88.17±8.91 |

#### IV. CONCLUSION

This paper mainly studies the intervention effect of 10 Hz electrical stimulation on the discriminability of left-hand and right-hand motor imagery tasks. After analyzing the EEG data in the pre- and post-intervention, we found that the activity in the sensory motor areas was significantly increased during motor imagination, and the left- and right-hand motor imagery tasks were easier to distinguish after intervention. Furthermore, we found that the classification accuracy of motor imagery tasks was improved after intervention.

The method of this paper is to improve BCI performance by activating the activity of contralateral cerebral cortex. Some researchers have studied that enhanced cortical activation in damaged motor cortex can be used as an indicator of motor deficit recovery [23,24]. Therefore, the experimental paradigm proposed in this paper could not only enhance the performance of BCI and improve the classification accuracy, but also play an auxiliary role in nerve repair of patients with motor disorders such as stroke. This work provides a new method to assist in the treatment of motor dyskinesia patients with somatic nerve repair. And the results of this study will also provide support for the rehabilitation research of patients with dyskinesia in next stage.

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